

Exercise 4: Unscrambling the Secret Codes

Goals:

To enable the team members to experience team problem-solving processes

To give the team members an opportunity to observe and identify behaviors and methods that facilitate or hinder effective teamwork.

To highlight the consequences of conflicts between individual objectives and team objectives

To provide a basis for exploring means to make teamwork more effective

Group Size:

8-16 members per subgroup

Time:

Approx. 1 hour

Material:

- A copy of the Unscrambling the Secret Codes Fact Sheet for each team
- A set of Unscrambling the Secret Codes Data Cards for each subgroup of the team
- Blank paper and a pencil for each team member
- A timing device
- A copy of the Unscrambling the Secret Codes Answer Sheet

Process:

- Divide the team into two subgroups of approx. equal size
- Each team member is given blank paper and a pencil
- Each subgroup is given a copy of the fact sheet and one set of the data cards; the cards are distributed as evenly as possible among the subgroup members. The subgroup members are told not to reveal the information of their cards to anyone else at this time.
- The CoC or cadet in charge instructs the teams to study the fact sheet and each member should study each of the cards assigned to them (5 minutes)
- The CoC or cadet in charge explains that the subgroups will be timed as they unscramble the secret codes and match the munitions with the appropriate base, secret code, yield and aircraft. The members of each subgroup are told that during the activity they may discuss the information on the cards that were assigned to them, but they may not pass the cards around for others to see. The CoC or cadet in charge also explains the scoring system, tells the subgroup members that no more questions will be answered, and instructs the subgroup members to raise their hands when they arrive at a solution.
- The CoC or cadet in charge starts the timing device and tells the teams to begin
- When a hand is raised, the CoC or cadet in charge makes a note of the time and then checks the answer for accuracy. If any part of the answer is wrong, the CoC or cadet in charge merely tells the subgroup member or the subgroup to continue working on the problem because the answer is not correct. (25 minutes)

- After both subgroups have found the correct solution, the CoC or cadet in charge leads a discussion based on the following questions:
 - What individual behaviors and problem-solving methods facilitated your subgroup in solving the problem? What individual behaviors and problem-solving methods hindered it?
 - At what points were you tempted to leave the subgroup and try to solve the problem on your own? What choice did you make? How do you account for your choice?
 - When an individual dropped out of your subgroup, how did you feel? How did you feel when the individual rejoined your subgroup?
 - What did you learn about conflict between individual objectives and team objectives and its effect on teamwork?
 - In what ways could you make the teamwork more effective in this team?

Unscrambling the Secret Codes Fact Sheet

Five nuclear munitions:

Blue bomb; green bomb; purple missile; red missile; yellow rocket

Five bases where one each of the nuclear munitions are located:

Alpha AFB; Bravo AFB; Charlie AFB; Delta AFB; Echo AFB

Five secret codes to use one each of the nuclear munitions:

727253; 1799351; 4219530; 10429538; 42911786

The yields of the five nuclear munitions (not in order):

10K; 15K; 20K; 25K; 30K

The aircraft that delivers one each of the five nuclear munitions (not in order):

B-99; FB-50; B-5; FB-12; A-25

Instructions

Your subgroup's task is to match each of the nuclear munitions with its base, secret code, yield, and the aircraft used to deliver it.

Scoring

If your subgroup has solved the problem correctly in every aspect the first time it submits an answer, it will receive a score of 100 minus the number of minutes it took to find the solution.

Each time a subgroup submits an answer that is not correct in every aspect, 5 points will be deducted from its score as it continues to try to solve the problem.

At any time, you – as an individual – may drop out of the subgroup effort and propose your individual solution. If the first individual answer that you submit is correct in every aspect, your score will be 100 minus half the number of minutes that were taken to solve the problem. You may then share the correct answer with your subgroup, and your individual score will become your subgroup's score.

If your solution is not correct in every aspect, you may rejoin the subgroup and deduct 10 points from the subgroup's score (that is, 100 minus the number of minutes required to solve the problem minus an additional 10 points). This will be your only opportunity to rejoin the subgroup.

If you choose to continue to work on your own, deduct 15 points from your individual score. For each additional time that you submit an incorrect answer, deduct 5 points from your individual score.

Unscrambling the Secret Codes Data Cards

Prior to conducting the activity, the CoC or cadet in charge must cut these statements into individual cards or strips. There must be one set per subgroup.

1. The yellow rocket's secret code and the green bomb's code contain the same number of digits
2. The red missile is delivered by the A-25
3. The green bomb is not delivered by the B-5
4. The yield of the nuclear munitions that's delivered by the B-99 is 15K
5. The secret code used at Charlie AFB has six digits
6. The yellow rocket is not located at Delta AFB
7. The yield of the nuclear munitions located at Alpha AFB IS 15k
8. The blue bomb is not located at Delta AFB
9. The nuclear munition's secret code delivered by the B-99 is 42911786
10. The yield of the nuclear munitions with the secret code 4219530 is 20K
11. The yield of the nuclear munitions that's delivered by the B-5 is less than 15K
12. The yellow rocket is delivered by the FB-12
13. The yield of the nuclear munitions located at Charlie AFB is more than 10K
14. The yield of the green bomb is 25K
15. The FB-12 is an underground hangar at Echo AFB
16. The blue bomb's secret code is 10429538

Unscrambling the Secret Codes Answer Sheet

<u>Nuclear Munitions</u>	<u>Aircraft</u>	<u>Base</u>	<u>Secret Codes</u>	<u>Yield</u>
Blue Bomb	B-5	Bravo AFB	10429538	10K
Green Bomb	FB-50	Delta AFB	1799351	25K
Purple Missile	B-99	Alpha AFB	42911786	15K
Red Missile	A-25	Charlie AFB	727253	30K
Yellow Rocket	FB-12	Echo AFB	42919530	20K